

```
# Skill3 - Data Analysis using pandas
# 1] Data Understanding & Loading
import pandas as pd
import numpy as np
```

```
df = pd.read_csv("movies - movies.csv")
```

```
df.head()
```

	name	rating	genre	year	released	score	votes	director
0	The Shining	R	Drama	1980	June 13, 1980 (United States)	8.4	927000.0	Stanley Kubrick
1	The Blue Lagoon	R	Adventure	1980	July 2, 1980 (United States)	5.8	65000.0	Randal Kleiser
2	Star Wars: Episode V - The Empire Strikes Back	PG	Action	1980	June 20, 1980 (United States)	8.7	1200000.0	Irvin Kershner
3	Airplane!	PG	Comedy	1980	July 2, 1980 (United States)	7.7	221000.0	Jim Abrahams
4	Caddyshack	R	Comedy	1980	July 25, 1980 (United States)	7.3	108000.0	Harold Ramis

```
df.tail()
```

	name	rating	genre	year	released	score	votes	director	writer
7663	More to Life	NaN	Drama	2020	October 23, 2020 (United States)	3.1	18.0	Joseph Ebanks	Joseph Eban
7664	Dream Round	NaN	Comedy	2020	February 7, 2020 (United States)	4.7	36.0	Dusty Dukatz	Lisa Huston
7665	Saving Mbango	NaN	Drama	2020	April 27, 2020 (Cameroon)	5.7	29.0	Nkanya Nkwai	Lynne Love
7666	It's Just Us	NaN	Drama	2020	October 1, 2020 (United States)	NaN	NaN	James Randall	James Rand
7667	Tee em el	NaN	Horror	2020	August 19, 2020 (United States)	5.7	7.0	Pereko Mosia	Pere Mos

```
df.shape
```

```
(7668, 15)
```

```
df.columns
```

```
Index(['name', 'rating', 'genre', 'year', 'released', 'score', 'votes',  
      'director', 'writer', 'star', 'country', 'budget', 'gross',  
      'company',  
      'runtime'],  
      dtype='object')
```

```
df.dtypes
```

```
name          object  
rating        object  
genre         object  
year          int64  
released      object  
score         float64  
votes         float64  
director      object  
writer        object  
star          object  
country       object  
budget        float64  
gross         float64  
company       object  
runtime       float64  
dtype: object
```

```
# Data Cleaning & Preprocessing
# a) Detect missing values
df.isnull().sum()
```

```
name          0
rating        77
genre         0
year          0
released      2
score         3
votes         3
director      0
writer        3
star          1
country       3
budget       2171
gross        189
company       17
runtime       4
dtype: int64
```

```
# b) Handling missing values
# Fill numerical missing values median
df['budget'] = df['budget'].fillna(df['budget'].median())
df['gross'] = df['gross'].fillna(df['gross'].median())

# Drop rows with missing critical values
df.dropna(subset=['rating', 'genre'], inplace=True)
```

```
# c) Remove duplicates
df.drop_duplicates(inplace=True)
```

```
# d) Rename Columns
df.rename(columns={'name': 'movie_name'}, inplace=True)
```

```
# e) Change data types
df['year'] = df['year'].astype(int)
```

```
# f) String Cleaning
df['genre'] = df['genre'].str.lower().str.strip()
df['country'] = df['country'].str.lower().str.strip()
```

```
# 3] Data Selection & Filtering
# a) Column selection
df[['movie_name', 'genre', 'score']]
```

	movie_name	genre	score
0	The Shining	drama	8.4
1	The Blue Lagoon	adventure	5.8
2	Star Wars: Episode V - The Empire Strikes Back	action	8.7
3	Airplane!	comedy	7.7
4	Caddyshack	comedy	7.3
...	...	...	...
7655	Legend of Deification	animation	6.6
7656	Tulsa	comedy	5.0
7658	Black Wall Street Burning	drama	6.6
7659	I Am Fear	horror	3.4
7661	Love by Drowning	drama	NaN

7591 rows × 3 columns

```
# b) Row selection
df.iloc[0:5]
```

	movie_name	rating	genre	year	released	score	votes	director
0	The Shining	R	drama	1980	June 13, 1980 (United States)	8.4	927000.0	Stanley Kubrick
1	The Blue Lagoon	R	adventure	1980	July 2, 1980 (United States)	5.8	65000.0	Randal Kleiser
2	Star Wars: Episode V - The Empire Strikes Back	PG	action	1980	June 20, 1980 (United States)	8.7	1200000.0	Irvin Kershner
3	Airplane!	PG	comedy	1980	July 2, 1980 (United States)	7.7	221000.0	Jim Abrahams
4	Caddyshack	R	comedy	1980	July 25, 1980 (United States)	7.3	108000.0	Harold Ramis

```
# c) loc[] and iloc[]
df.loc[df['genre'] == 'comedy']
```

	movie_name	rating	genre	year	released	score	votes	director
3	Airplane!	PG	comedy	1980	July 2, 1980 (United States)	7.7	221000.0	Jim Abrahams
4	Caddyshack	R	comedy	1980	July 25, 1980 (United States)	7.3	108000.0	Harold Ramis
17	9 to 5	PG	comedy	1980	December 19, 1980 (United States)	6.9	29000.0	Colin Higgins
19	Stir Crazy	R	comedy	1980	December 12, 1980 (United States)	6.8	26000.0	Sidney Poitier
26	Little Darlings	R	comedy	1980	March 21, 1980 (United States)	6.5	5100.0	Ron Maxwell
...	...	...	...	...	...	...	...	...
7620	Shed of the Dead	R	comedy	2019	May 20, 2019 (United Kingdom)	4.7	1400.0	Drew Cullingham
7621	The Fighting Preacher	PG	comedy	2019	July 24, 2019 (United States)	5.9	403.0	T.C. Christensen
7626	No Manches Frida 2	R	comedy	2019	March 15, 2019 (United States)	4.2	904.0	Nacho G. Velilla
7633	Tod@s Caen	PG-13	comedy	2019	August 30, 2019 (United States)	4.9	550.0	Ariel Winograd
7656	Tulsa	PG-13	comedy	2020	June 3, 2020 (United States)	5.0	294.0	Scott Pryor

2233 rows × 15 columns

```
# d) Conditional filtering
df[df['score'] > 8]
```

	movie_name	rating	genre	year	released	score	votes	direct
0	The Shining	R	drama	1980	June 13, 1980 (United States)	8.4	927000.0	Stan Kubr
2	Star Wars: Episode V - The Empire Strikes Back	PG	action	1980	June 20, 1980 (United States)	8.7	1200000.0	Ir Kersh
7	Raging Bull	R	biography	1980	December 19, 1980 (United States)	8.2	330000.0	Mar Scorse
49	Lion of the Desert	PG	biography	1980	May 16, 1981 (Japan)	8.3	15000.0	Moustap Akk
66	Moscow Does Not Believe in Tears	PG	comedy	1980	February 11, 1980 (Soviet Union)	8.1	12000.0	Vladim Mensh
...	...	...	...	...	...	...	...	...
7462	1917	R	drama	2019	January 10, 2020 (United States)	8.3	473000.0	Sa Menc
7464	Ford v Ferrari	PG-13	action	2019	November 15, 2019 (United States)	8.1	323000.0	Jam Mang
7487	Portrait of a Lady on Fire	R	drama	2019	February 14, 2020 (United States)	8.1	74000.0	Céli Sciam
7555	Chhichhore	Not Rated	comedy	2019	September 6, 2019 (United States)	8.3	40000.0	Nite Tiw
7586	Sonchiriya	Not Rated	action	2019	March 1, 2019 (India)	8.1	9800.0	Abhish Chaut

189 rows × 15 columns

```
# e) Multiple conditions
df[(df['genre'] == 'action') & (df['score'] > 7)]
```

	movie_name	rating	genre	year	released	score	votes	director
2	Star Wars: Episode V - The Empire Strikes Back	PG	action	1980	June 20, 1980 (United States)	8.7	1200000.0	Irvin Kershner
6	The Blues Brothers	R	action	1980	June 20, 1980 (United States)	7.9	188000.0	John Landis
42	The Stunt Man	R	action	1980	June 27, 1980 (United States)	7.1	9000.0	Richard Rush
92	Indiana Jones and the Raiders of the Lost Ark	PG	action	1981	June 12, 1981 (United States)	8.4	905000.0	Steven Spielberg
95	Escape from New York	R	action	1981	July 10, 1981 (United States)	7.2	131000.0	John Carpenter
...	...	...	...	...	...	...	...	...
7468	Alita: Battle Angel	PG-13	action	2019	February 14, 2019 (United States)	7.3	240000.0	Robert Rodriguez
7581	Extreme Job	Not Rated	action	2019	June 21, 2019 (United States)	7.1	7600.0	Byeong- heon Lee
7586	Sonchiriya	Not Rated	action	2019	March 1, 2019 (India)	8.1	9800.0	Abhishek Chaubey
7589	Batla House	Not Rated	action	2019	August 15, 2019 (United States)	7.2	10000.0	Nikkhil Advani
7643	Tenet	PG-13	action	2020	September 3, 2020 (United States)	7.4	387000.0	Christopher Nolan

314 rows × 15 columns

```
# 4] Data Transformation & Feature Engineering
# a) Creating new columns
df['profit'] = df['gross'] - df['budget']
```

```
# b) Mathematical operations
df['budget_million'] = df['budget'] / 1_000_000
```

```
# c) c) Applying functions
df['hit_movie'] = df['score'].apply(lambda x: 'Hit' if x >= 7 else 'Average')
```

```
# d) Binning data
df['score_category'] = pd.cut(
    df['score'],
    bins=[0,5,7,10],
    labels=['Low', 'Medium', 'High']
)
```

```
# 5] Aggregation & Group-Based Analysis
# a) groupby()
df.groupby('genre')['score'].mean()
```

```
genre
action      6.202354
adventure    6.286854
animation    6.758209
biography    7.033864
comedy       6.192790
crime        6.671560
drama        6.691633
family       6.540000
fantasy      6.048837
horror       5.751887
music        7.200000
musical      7.900000
mystery      6.665000
romance      6.500000
sci-fi       6.266667
sport        5.900000
thriller     5.912500
western      5.666667
Name: score, dtype: float64
```

```
# b) Aggregation functions
df.groupby('genre').agg({
    'budget': 'mean',
    'gross': 'mean',
    'score': 'mean'
})
```


	budget	gross	score
genre			
action	5.209083e+07	1.433839e+08	6.202354
adventure	3.996845e+07	1.080905e+08	6.286854
animation	6.660170e+07	2.393742e+08	6.758209
biography	2.394518e+07	4.756948e+07	7.033864
comedy	2.193340e+07	4.394812e+07	6.192790
crime	2.190412e+07	3.944325e+07	6.671560
drama	2.199836e+07	3.854556e+07	6.691633
family	3.275000e+07	2.157876e+08	6.540000
fantasy	1.737674e+07	3.880865e+07	6.048837
horror	1.430296e+07	4.662030e+07	5.751887
music	2.050000e+07	1.100140e+05	7.200000
musical	2.050000e+07	2.217255e+06	7.900000
mystery	3.017000e+07	1.011835e+08	6.665000
romance	2.271250e+07	2.354937e+07	6.500000
sci-fi	2.205556e+07	3.118840e+07	6.266667
sport	2.050000e+07	1.067629e+06	5.900000
thriller	1.488125e+07	2.525288e+07	5.912500
western	1.383333e+07	1.067530e+07	5.666667

```
# c) Multiple aggregations
df.groupby('country')['movie_name'].count()
```

```
country
argentina      8
aruba           1
australia     91
austria        5
belgium        8
brazil         6
canada       187
chile          2
china         36
colombia       1
czech republic 8
denmark       31
federal republic of yugoslavia 2
finland        3
france       265
germany       114
greece         2
```

hong kong	43
hungary	4
iceland	2
india	61
indonesia	2
iran	10
ireland	43
israel	5
italy	58
jamaica	1
japan	75
kenya	1
lebanon	1
libya	1
malta	1
mexico	21
netherlands	12
new zealand	25
norway	12
panama	1
philippines	1
poland	4
portugal	2
republic of macedonia	1
romania	1
ruissia	8
serbia	1
south africa	7
south korea	31
soviet union	2
spain	47
sweden	24
switzerland	10
taiwan	7
thailand	6
turkey	3
united arab emirates	2
united kingdom	810
united states	5457

```
# d) pivot_table()
pd.pivot_table(
    df,
    values='gross',
    index='genre',
    columns='rating',
    aggfunc='mean'
)
```

rating	Approved	G	NC-17	Not Rated	PG
genre					
action	NaN	3.432739e+07	NaN	5.358252e+07	1.040077e+08
adventure	36565280.0	5.623710e+07	NaN	2.760398e+07	1.142305e+08
animation	NaN	1.858594e+08	NaN	8.134290e+07	3.143682e+08
biography	NaN	6.416569e+06	2.347245e+07	1.255054e+07	4.546726e+07
comedy	NaN	1.001722e+08	1.702544e+06	8.706993e+06	4.501870e+07
crime	NaN	NaN	1.500800e+07	7.121391e+06	9.749639e+06
drama	NaN	1.670851e+07	1.209224e+07	6.281319e+06	3.600288e+07
family	NaN	5.526314e+06	NaN	NaN	3.058996e+08
fantasy	NaN	NaN	NaN	1.957475e+05	1.367703e+07
horror	NaN	NaN	2.956630e+06	1.759841e+06	2.192766e+07
music	NaN	NaN	NaN	NaN	NaN
musical	NaN	NaN	NaN	2.217255e+06	NaN
mystery	NaN	NaN	NaN	4.592340e+05	NaN
romance	NaN	NaN	NaN	6.465320e+05	1.140285e+07
sci-fi	NaN	NaN	NaN	2.244000e+04	1.521261e+07
sport	NaN	NaN	NaN	NaN	NaN
thriller	NaN	NaN	NaN	NaN	9.032747e+05
western	NaN	NaN	NaN	NaN	1.736123e+06

```
# 6] Sorting, Ranking & Top-N Analysis
# a) sort_values()
df.sort_values(by='gross', ascending=False).head(10)
```

	movie_name	rating	genre	year	released	score	votes	director
5445	Avatar	PG-13	action	2009	December 18, 2009 (United States)	7.8	1100000.0	James Cameron
7445	Avengers: Endgame	PG-13	action	2019	April 26, 2019 (United States)	8.4	903000.0	Anthony Russo
3045	Titanic	PG-13	drama	1997	December 19, 1997 (United States)	7.8	1100000.0	James Cameron
6663	Star Wars: Episode VII - The Force Awakens	PG-13	action	2015	December 18, 2015 (United States)	7.8	876000.0	J.J. Abrams
7244	Avengers: Infinity War	PG-13	action	2018	April 27, 2018 (United States)	8.4	897000.0	Anthony Russo
7480	The Lion King	PG	animation	2019	July 19, 2019 (United States)	6.9	222000.0	Jon Favreau
6653	Jurassic World	PG-13	action	2015	June 12, 2015 (United States)	7.0	593000.0	Colin Trevorrow
6043	The Avengers	PG-13	action	2012	May 4, 2012 (United States)	8.0	1300000.0	Jos Whedon
6646	Furious 7	PG-13	action	2015	April 3, 2015 (United States)	7.1	370000.0	James Wan
7494	Frozen II	PG	animation	2019	November 22, 2019 (United States)	6.8	148000.0	Chris Buck

```
# b) nlargest(), nsmallest()
df.nlargest(5, 'score')[['movie_name', 'score']]
```

	movie_name	score
2443	The Shawshank Redemption	9.3
5243	The Dark Knight	9.0
2247	Schindler's List	8.9
2444	Pulp Fiction	8.9
4245	The Lord of the Rings: The Return of the King	8.9

```
# c) Ranking
df['score_rank'] = df['score'].rank(ascending=False)
```

```
# 7] Data Merging & Combining
# Create a sample dataframe
ratings_df = df[['movie_name', 'rating']]

# Merge example
merged_df = pd.merge(df, ratings_df, on='movie_name', how='inner')
merged_df.head()
```

	movie_name	rating_x	genre	year	released	score	votes	director
0	The Shining	R	drama	1980	June 13, 1980 (United States)	8.4	927000.0	Stanley Kubrick
1	The Blue Lagoon	R	adventure	1980	July 2, 1980 (United States)	5.8	65000.0	Randal Kleiser
2	Star Wars: Episode V - The Empire Strikes Back	PG	action	1980	June 20, 1980 (United States)	8.7	1200000.0	Irvin Kershner
3	Airplane!	PG	comedy	1980	July 2, 1980 (United States)	7.7	221000.0	Jim Abrahams
4	Caddyshack	R	comedy	1980	July 25, 1980 (United States)	7.3	108000.0	Harold Ramis

5 rows × 21 columns

```
# 8] Time-Series Handling
# a) Date parsing
```

```
df['released_clean'] = df['released'].str.extract(r'(\w+ \d{1,2}, \d{4})')  
df['released_date'] = pd.to_datetime(df['released_clean'], format='%B %d, %Y')
```

```
# b) Extract year, month, day  
df['release_year'] = df['released_date'].dt.year  
df['release_month'] = df['released_date'].dt.month
```

```
# c) Sorting by date  
df.sort_values(by='released_date')
```

	movie_name	rating	genre	year	released	score	votes	director
81	Windows	R	drama	1980	January 18, 1980 (United States)	4.8	643.0	Gordon Willis
74	Just Tell Me What You Want	R	comedy	1980	January 18, 1980 (United States)	5.5	585.0	Sidney Lumet
29	American Gigolo	R	crime	1980	February 1, 1980 (United States)	6.2	22000.0	Paul Schrader
65	Fatso	PG	comedy	1980	February 1, 1980 (United States)	6.3	2000.0	Anne Bancroft
18	The Fog	R	horror	1980	February 8, 1980 (United States)	6.8	66000.0	John Carpenter
...	...	...	...	...	...	...	...	...
5728	Saw: The Final Chapter	R	crime	2010	NaN	5.6	93000.0	Kevin Greutert
5730	The Wolfman	R	drama	2010	NaN	5.8	104000.0	Joe Johnston
5833	Hatchet II	R	action	2010	November 2010 (Australia)	5.5	13000.0	Adam Green
5958	The Human Centipede II (Full Sequence)	Not Rated	horror	2011	November 2011 (Australia)	3.8	37000.0	Tom Six
6413	Romeo and Juliet	PG-13	drama	2013	2013 (United States)	5.8	11000.0	Carlo Carlei

7591 rows × 24 columns

```
# d) Time-based filtering
df[df['release_year'] > 2010]
```

	movie_name	rating	genre	year	released	score	votes	director
1915	Only Yesterday	PG	animation	1991	February 26, 2016 (United States)	7.6	29000.0	Isao Takahata
2219	Hyenas	Not Rated	comedy	1992	April 26, 2019 (United States)	7.4	1000.0	Djibril Diop Mambéty
2974	A Summer's Tale	G	comedy	1996	June 20, 2014 (United States)	7.6	7000.0	Éric Rohmer
4212	No Good Deed	R	crime	2002	November 20, 2014 (Germany)	5.4	7100.0	Bob Rafelson
5106	The Babysitters	R	drama	2007	May 13, 2011 (Poland)	5.6	11000.0	David Ross
...	...	...	...	...	...	...	...	...
7655	Legend of Deification	TV-PG	animation	2020	October 1, 2020 (United States)	6.6	1300.0	Teng Cheng
7656	Tulsa	PG-13	comedy	2020	June 3, 2020 (United States)	5.0	294.0	Scott Pryor
7658	Black Wall Street Burning	R	drama	2020	February 7, 2020 (United States)	6.6	35.0	Marcus Brown
7659	I Am Fear	Not Rated	horror	2020	March 3, 2020 (United States)	3.4	447.0	Kevin Shulman
7661	Love by Drowning	R	drama	2020	November 6, 2020 (United States)	NaN	NaN	Justin Kreinbrink

1844 rows × 24 columns

```
# 9] Exploratory Data Analysis (EDA)
# a) Summary statistics
df.describe()
```


	year	score	votes	budget	gross	
count	7591.000000	7590.000000	7.590000e+03	7.591000e+03	7.591000e+03	7588
mean	2000.372283	6.388050	8.878900e+04	3.146656e+07	7.761253e+07	107
min	1980.000000	1.900000	3.500000e+01	3.000000e+03	3.090000e+02	63
25%	1991.000000	5.800000	9.300000e+03	1.400000e+07	4.792905e+06	95
50%	2000.000000	6.500000	3.300000e+04	2.050000e+07	2.020576e+07	104
75%	2010.000000	7.100000	9.400000e+04	3.200000e+07	7.394282e+07	116
max	2020.000000	9.300000	2.400000e+06	3.560000e+08	2.847246e+09	366
std	11.116688	0.968424	1.639090e+05	3.588963e+07	1.645060e+08	18

```
# b) Value counts
df['genre'].value_counts()
```

```
genre
comedy      2233
action      1699
drama       1483
crime        545
biography   440
adventure   426
animation   335
horror       318
fantasy      43
mystery      20
thriller     16
family       10
sci-fi        9
romance       8
western       3
music         1
sport         1
musical       1
Name: count, dtype: int64
```

```
# c) Correlation
df[['budget', 'gross', 'score']].corr()
```

budget gross score									
# d) Detecting outliers df[df['gross'] > df['gross'].quantile(0.99)]									
score	movie	year	released	score	votes	director			
2245	Jurassic Park	1993	June 11, 1993 (United States)	8.1	894000.0	Steve Spielberg			
2448	The Lion King	1994	June 24, 1994 (United States)	8.5	970000.0	Roger Allers			
3045	Titanic	1997	December 19, 1997 (United States)	7.8	1100000.0	James Cameron			
3451	Star Wars: Episode I - The Phantom Menace	1999	May 19, 1999 (United States)	6.5	748000.0	George Lucas			
3843	Harry Potter and the Sorcerer's Stone	2001	November 16, 2001 (United States)	7.6	685000.0	Chris Columbus			
...	...	...	...	...	...	...			
7463	Star Wars: Episode IX - The Rise of Skywalker	2019	December 20, 2019 (United States)	6.5	394000.0	J.J. Abrams			
7480	The Lion King	2019	July 19, 2019 (United States)	6.9	222000.0	Jon Favreau			
7481	Aladdin	2019	May 24, 2019 (United States)	6.9	239000.0	Guy Ritchie			